HDOC Day-67

Question

Write a recursive function to find the GCD of two positive integers using Euclid's algorithm.

Python Program

```
def gcd(a, b):
    if b == 0:
        return a
    return gcd(b, a % b)

num1 = int(input("Enter first positive integer: "))
num2 = int(input("Enter second positive integer: "))

if num1 > 0 and num2 > 0:
    print("GCD =", gcd(num1, num2))
else:
    print("Please enter positive integers.")
```

Sample Output

```
Enter first positive integer: 24
Enter second positive integer: 36
GCD = 12
```

Explanation

Euclid's algorithm repeatedly replaces the pair (a, b) with (b, a % b). The recursive function stops when b becomes 0. At that point, a is the greatest common divisor (GCD).